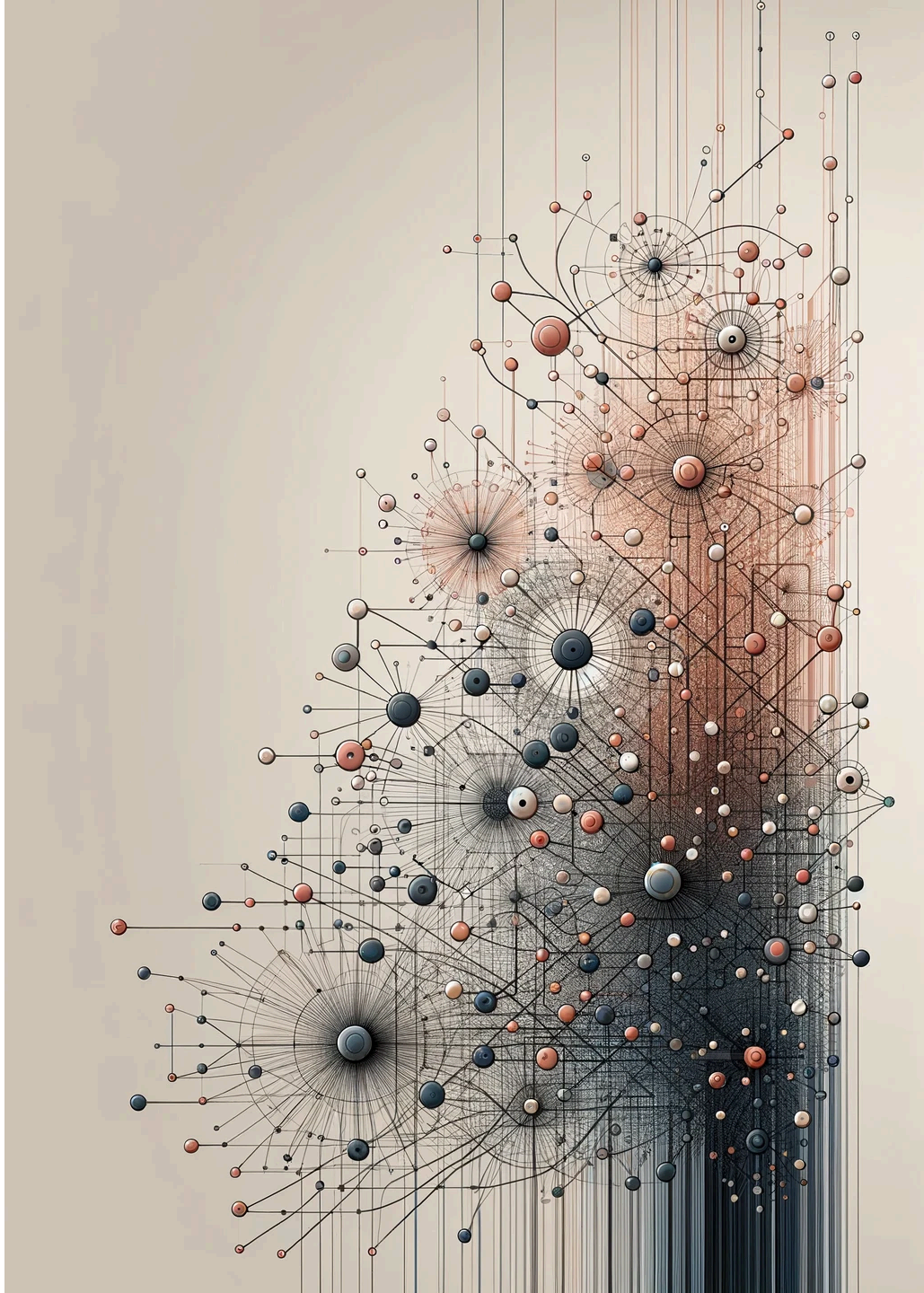
UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS09 - Preferential Attachment

Prof. Rossano Schifanella



Outline

- Why static models are not enough
- Growth and preferential attachment
- The Barabási-Albert model
- What BA explains
- Why linearity is special
- What BA still misses

From static models to growth models

In `NS08` we studied models that explain:

- random connectivity
- shortcut creation
- degree-preserving randomization

They are useful baselines, but they do not explain how **hubs emerge through network growth**.

The central modeling question

Many real networks are not built all at once.

They grow over time, and new nodes do not connect uniformly at random.

So the key question is:

> Which minimal growth rule can generate hubs rather than impose them by hand?

Growth is a structural mechanism

Real networks are dynamic.

Examples:

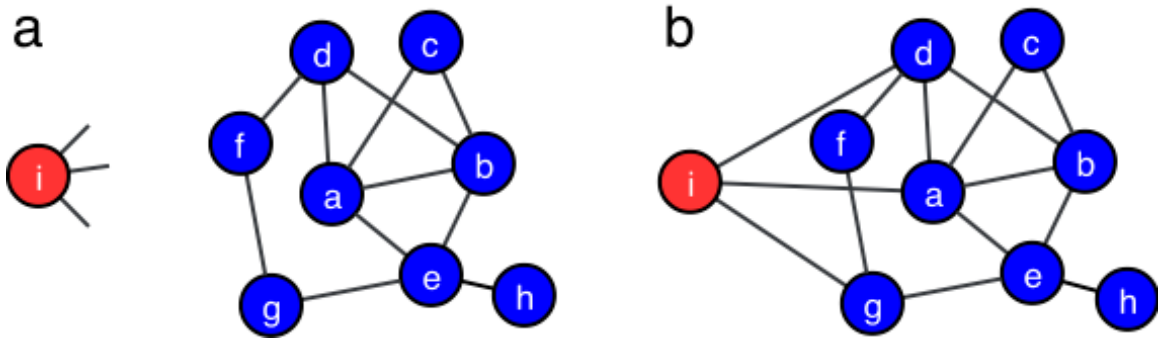
- the Web grows as new pages appear
- citation networks grow as new papers are published
- collaboration networks grow as new authors and papers enter the system
- technological and biological networks also change over time

Growth is therefore not a detail. It is part of the mechanism.

Growth model: generic template

A generic growth model has two ingredients:

1. new nodes arrive over time
2. each new node chooses where to connect according to some rule



The key modeling choice is the attachment rule.

Preferential attachment: the idea

Preferential attachment means:

- highly connected nodes are more likely to receive new links
- connectivity creates further visibility, exposure, or opportunity
- advantage accumulates over time

This is a network version of **cumulative advantage**.

Why preferential attachment is plausible

Examples:

- on the Web, popular pages are easier to discover and therefore easier to link to
- highly cited papers are more visible and more likely to be cited again
- actors or creators with many connections tend to receive more opportunities

The assumption is not that degree is the only factor.

The assumption is that degree can influence future attachment.

A useful analogue: Polya's urn

Start with an urn containing balls of different colors.

At each step:

1. draw one ball
2. return it to the urn
3. add another ball of the same color

A color that is already frequent becomes even more likely to be drawn again.

Why Polya's urn matters here

The urn model shows the logic of reinforcement:

- present success changes future probability
- small early differences can be amplified
- the process produces unequal outcomes without central coordination

Preferential attachment applies the same idea to link formation in growing networks.

What a hub-generating model must contain

A minimal model for hub emergence needs two ingredients:

- **growth**: the number of nodes increases over time
- **preferential attachment**: new links favor already well-connected nodes

Growth alone is not enough.

Preferential attachment without growth is also not enough.

The Barabási-Albert model

The Barabási-Albert model: definition

Start from an initial graph with m_0 nodes.

At each step:

1. add a new node i
2. create m new links from i to existing nodes, with $m \leq m_0$
3. choose each target node j with probability proportional to its degree

$$P(i \leftrightarrow j) = \frac{k_j}{\sum_l k_l}$$

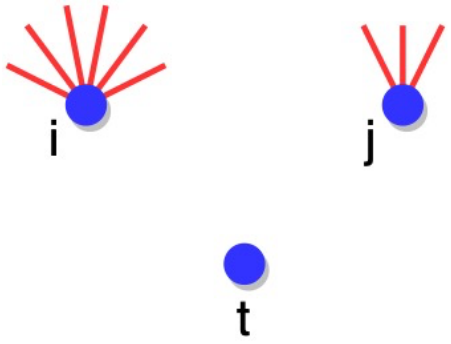
Reading the attachment probability

If node a has degree 6 and node b has degree 3, then under linear preferential attachment:

$$P(i \leftrightarrow a) = 2\,P(i \leftrightarrow b)$$

The model does not choose uniformly.

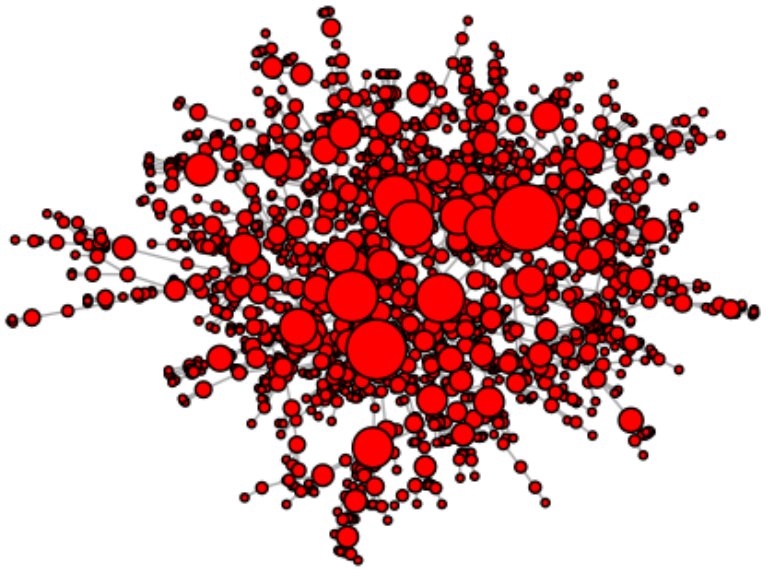
It weights attachment opportunities by current connectivity.



What the BA mechanism does

Preferential attachment creates a **rich-get-richer** effect:

- nodes with larger degree receive more opportunities
- those extra links increase their future advantage
- hubs emerge endogenously from the growth process



Is growth alone enough?

Older nodes usually have more time to collect links.

So we should ask a sharper question:

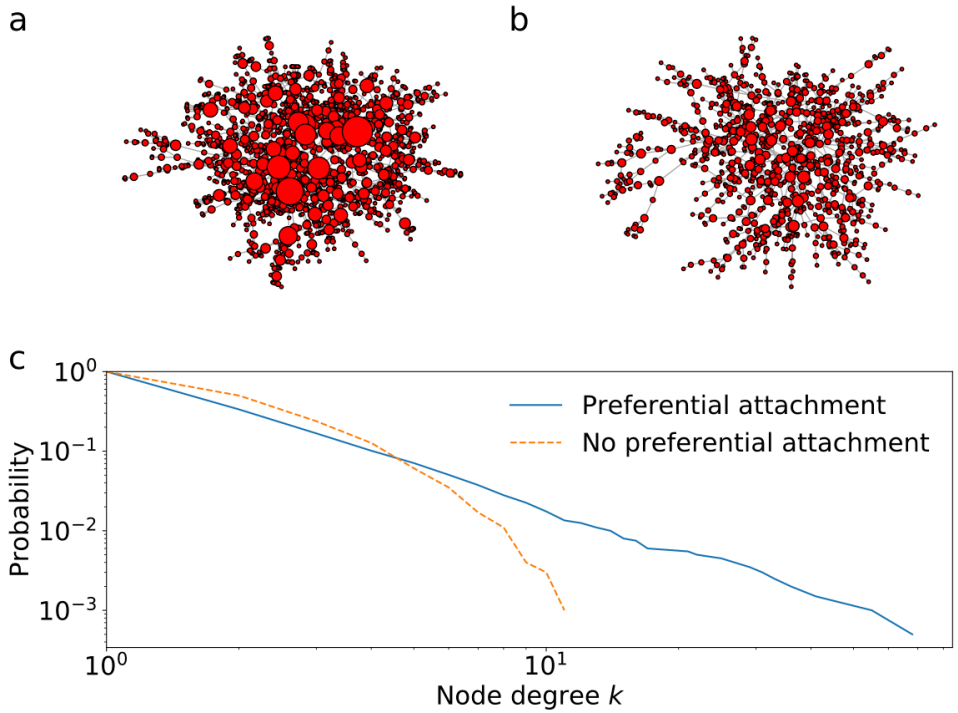
> If nodes arrive over time but targets are chosen uniformly at random, do hubs still emerge?

The answer is no.

Growth without preferential attachment does not create hubs

Conclusion: growth plus uniform attachment does **not** generate the broad hub structure observed in scale-free networks.

Preferential attachment is the critical ingredient.



Why preferential attachment matters structurally

Preferential attachment changes the shape of the degree distribution.

Instead of a narrow distribution centered around an average degree, the model creates a broad hierarchy:

- many low-degree nodes
- a small number of very highly connected nodes

This is the structural signature of hub emergence.

Heuristic derivation of the BA exponent

The exact BA process is random:

- at each step, a node may or may not receive a new link
- degree grows through discrete jumps, not smoothly

To derive a simple formula, we replace those random jumps with their average effect over time.

So $k_i(t)$ is treated as a smooth approximation of the expected degree of node i.

What the derivation is approximating

At time t, the network has about mt edges, because each arriving node brings m new links.

Therefore the total degree is

$$\sum_l k_l \approx 2mt$$

Each edge contributes 2 to the total degree, one for each endpoint.

This is the normalization term in the attachment probability.

Step 1: expected degree growth of one node

Consider node i, born at time t_i.

At each step, m new links are added.

Each one chooses node i with probability

$$\frac{k_i}{\sum_l k_l} \approx \frac{k_i}{2mt}$$

So the expected rate of increase of k_i is

$$\frac{dk_i}{dt} = m\frac{k_i}{2mt} = \frac{k_i}{2t}$$

There are m chances to receive a link at each step, and each chance is weighted by current degree.

The richer a node already is, the faster it grows.

Step 2: solve the growth equation

Starting from

$$\frac{dk_i}{dt} = \frac{k_i}{2t}$$

separate variables:

$$\frac{dk_i}{k_i} = \frac{dt}{2t}$$

All terms involving k_i are now on the left, and all terms involving t are on the right.

Integrate both sides:

$$\int \frac{1}{k_i}\, dk_i = \int \frac{1}{2t}\, dt$$

The logarithms appear because the integral of $1/x$ is $\ln x$.

This gives

$$\ln k_i + C_1 = \frac{1}{2} \ln t + C_2$$

Only the difference between the two constants matters, so they can be combined into a single constant:

$$\ln k_i = \frac{1}{2} \ln t + C$$

Instead of carrying two arbitrary constants, it is enough to keep one.

Apply $e^{(\cdot)}$ to both sides:

$$e^{\ln k_i} = e^{\frac{1}{2}\ln t + C}$$

Applying the exponential is the inverse operation of taking the logarithm.

Use $e^{a+b} = e^a e^b$:

$$e^{\ln k_i} = e^C \, e^{\frac{1}{2}\ln t}$$

Now simplify each term:

$$e^{\ln k_i} = k_i \qquad \text{and} \qquad e^{\frac{1}{2}\ln t} = t^{1/2}$$

Therefore

$$k_i(t) = e^C t^{1/2} = A t^{1/2}$$

So the solution has the form "constant times $t^{1/2}$", and the next step determines the constant.

Step 3: use the initial condition

The differential equation gives a family of possible curves. The initial condition selects the one corresponding to node i.

Node i enters the network at time t_i and arrives with exactly m links, so

$$k_i(t_i) = m$$

This says that every node starts with degree m at the moment it enters.

Substitute $t = t_i$ into the general solution:

$$m = At_i^{1/2}$$

Therefore

$$A = \frac{m}{t_i^{1/2}}$$

and

$$k_i(t) = \frac{m}{t_i^{1/2}} t^{1/2} = m \left(\frac{t}{t_i} \right)^{1/2}$$

This is the key growth law: at the same observation time t, older nodes tend to have larger degree.

Step 4: solve for the birth time

From

$$k_i(t) = m \left(\frac{t}{t_i} \right)^{1/2}$$

This formula gives the degree of node i at observation time t.

For a fixed observation time t, only nodes with $t_i \leq t$ already exist.

Among those nodes, the formula shows how the degree depends on the birth time t_i:

- smaller t_i means the node entered earlier
- earlier nodes have had more time to accumulate links

Now fix the observation time t and ask the inverse question: if a node has degree k at time t, how early must it have entered?

So we solve for t_i step by step.

Set $k_i(t) = k$, then divide by m:

$$\frac{k}{m} = \left(\frac{t}{t_i}\right)^{1/2}$$

Now the unknown is t_i, not k.

Then square both sides:

$$\left(\frac{k}{m}\right)^2 = \frac{t}{t_i}$$

Now isolate t_i:

$$t_i = \frac{tm^2}{k^2}$$

Equivalently,

$$t_i = m^2 t\, k^{-2}$$

Why this rewrite is useful

The formula

$$t_i = m^2 t\, k^{-2}$$

does not give the degree directly.

It gives a **birth-time threshold**:

- to have degree at least k at time t, a node must have entered early enough
- nodes born after that threshold are too young to have reached degree k

This is useful because birth times are easy to count in BA: one node arrives at each time step.

The whole purpose of this rewrite is to replace a hard degree-counting problem with an easy counting-by-age problem.

Interpreting the threshold

The condition

$$t_i \leq m^2 t\, k^{-2}$$

means:

- to have degree at least k by time t, a node must have entered before time $m^2 t\, k^{-2}$
- if a node entered later than that, it is too young to have reached degree k

So a degree condition becomes a birth-time condition.

Step 5: from birth time to the degree tail

A node has degree at least k if

$$k_i(t) \geq k \quad \Longleftrightarrow \quad t_i \leq m^2 t\, k^{-2}$$

At time t, the network has about t nodes, and one node was added at each step.

So the number of nodes with degree at least k is approximately the number of nodes born before time $m^2 t\, k^{-2}$, namely about $m^2 t\, k^{-2}$ nodes.

Therefore, in the tail regime, the fraction of nodes with degree at least k is

$$\frac{m^2 t\, k^{-2}}{t} \propto k^{-2}$$

The factor t cancels out, so the scaling with k is the important part.

Therefore

$$P(K \geq k) \propto k^{-2}$$

This is the quantity we will convert next into the degree distribution $p(k)$.

Step 6: why does the tail appear first?

The quantity

$$P(K \geq k)$$

answers a threshold question:

- what fraction of nodes have degree at least k?

It sums all probabilities from k upward:

$$P(K \geq k) = \sum_{q=k}^{\infty} p(q)$$

So it is called the **tail distribution**, because it measures the right tail of the degree distribution.

It is also called the **complementary cumulative distribution** because it is the complement of the usual cumulative distribution:

$$P(K \geq k) = 1 - P(K \leq k - 1)$$

What we ultimately want, however, is

$$p(k)$$

the fraction of nodes with degree exactly k.

The tail gives "at least k"; the degree distribution gives "exactly k".

Step 7: from the cumulative tail to $p(k)$

For a discrete variable such as degree,

$$p(k) = P(K \geq k) - P(K \geq k+1)$$

Why does this work?

- $P(K \geq k)$ counts nodes with degree $k, k+1, k+2, \ldots$
- $P(K \geq k+1)$ counts nodes with degree $k+1, k+2, \ldots$
- subtracting the second from the first leaves only the nodes with degree exactly k

Step 7b: why a derivative appears

In the continuous approximation, the change from threshold k to threshold $k+1$ is approximated by the slope of the tail curve.

So

$$P(K \geq k+1) - P(K \geq k) \approx \frac{d}{dk} P(K \geq k)$$

The tail decreases as k increases, so this derivative is negative.

Therefore

$$p(k) = P(K \geq k) - P(K \geq k+1) \approx -\frac{d}{dk} P(K \geq k)$$

If

$$P(K \geq k) \propto k^{-2},$$

then

$$p(k) \approx -\frac{d}{dk}k^{-2} = 2k^{-3} \propto k^{-3}$$

Proof recap

The logic of the derivation is:

1. average-growth approximation gives

$$\frac{dk_i}{dt} = \frac{k_i}{2t}$$

2. solving gives

$$k_i(t) = m\left(\frac{t}{t_i}\right)^{1/2}$$

3. large degree means small birth time, so a degree threshold becomes a birth-time threshold
4. counting early-born nodes gives

$$P(K \geq k) \propto k^{-2}$$

5. converting the tail into the exact degree distribution gives

$$p(k) \propto k^{-3}$$

So the BA model has the classical power-law exponent

$$\gamma = 3$$

The constant factor 2 and the minus sign do not matter for the exponent. What matters is the scaling with k.

Main property 1: a power-law degree distribution

The BA model generates

$$p(k) \sim k^{-3}$$

So the degree distribution is broad and heavy-tailed, unlike in Erdős-Rényi or Watts-Strogatz networks.

This is the main reason the model became central in early network science.

What this exponent means

The BA model does not just generate hubs.

It generates a specific scaling regime:

- the tail decays as a power law
- the exponent is fixed at $\gamma = 3$
- heterogeneity is therefore strong, but not arbitrary

So BA is a minimal explanation, not a flexible fit to every dataset.

Main property 2: short paths

Because hubs act as structural shortcuts, BA networks have short path lengths.

A classical scaling result is

$$\ell \sim \frac{\ln N}{\ln \ln N}$$

which is even smaller than the logarithmic scaling typical of homogeneous random graphs.

Implication of the path-length result

Once hubs emerge, they compress distances strongly.

So the same mechanism that creates broad degree heterogeneity also changes global navigability and spreading behavior.

Hub formation is therefore not just a local degree effect.

It changes the whole geometry of the network.

Main property 3: clustering remains low

The BA model reproduces hubs, but it does not reproduce strong local closure.

A classical asymptotic result is

$$C_{BA}(N) \sim \frac{m}{8} \frac{(\ln N)^2}{N} \to 0$$

So clustering vanishes as the network grows.

Why clustering stays low in BA

The model connects new nodes to important nodes, but it does not explicitly create triangles.

Therefore:

- hubs appear
- paths become short
- local closure is weak

This is the first major empirical limitation of the minimal BA mechanism.

Why linearity is special

The BA model uses **linear** preferential attachment:

$$P(i \leftrightarrow j) \propto k_j$$

A natural generalization is

$$P(i \leftrightarrow j) \propto k_j^{\alpha}$$

The behavior changes qualitatively when $\alpha \neq 1$.

Sublinear and superlinear attachment

- if $\alpha < 1$, high-degree nodes are not favored strongly enough
  - the degree distribution becomes less heterogeneous
  - large hubs weaken
- if $\alpha > 1$, reinforcement becomes too strong
  - one or a few nodes capture a macroscopic share of the links
  - a winner-takes-all regime appears

Conclusion: linear preferential attachment is the critical case

The classical scale-free BA regime appears at the linear point:

$$\alpha = 1$$

So linear preferential attachment is not just a convenient assumption.

It is the special case that balances reinforcement strongly enough to create hubs, but not so strongly as to collapse into condensation.

What BA explains well

- hub emergence from a simple local rule
- broad degree heterogeneity
- very short paths
- a clean connection between growth and cumulative advantage

These are major theoretical achievements for a minimal model.

What BA does not explain

- high clustering and triangle formation
- age-independent success of newer nodes
- attachment based on local search rather than global degree knowledge
- flexibility in degree exponents
- deletion, rewiring, and multiple components

These are not minor details. They are the main directions for model extension.

BA as a minimal model

The right way to read the BA model is:

- not as a full description of real networks
- not as proof that every broad degree distribution comes from preferential attachment
- but as the minimal generative model showing how hubs can emerge through growth and reinforcement

Summary

Question	BA answer
How do hubs emerge?	Growth + linear preferential attachment
What degree pattern appears?	Power law with $\gamma = 3$
What global property follows?	Very short paths
Main missing feature?	High clustering and richer attachment mechanisms

Next step: fixing BA's empirical failures

The BA model explains how hubs can emerge.

The next question is how to modify that minimal mechanism when real networks also require:

- baseline attractiveness
- quality differences across nodes
- local search and triadic closure
- copying and imitation mechanisms

Reading material

References

[ns1] **Chapter 5 (Network Models)**

Q&A